

# FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

*Teacher Key: Exemplar Answers and Marking Guide*

## FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10 – SUB-STRAND 2.2 REFLECTION: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

### INSTRUCTIONS GIVEN TO STUDENTS

Driving question: Why does a mirror show a perfect copy of the world, and where exactly does that copy live? Use everything you have learned from Lesson 1 to Lesson 8 to answer the questions below. Write your answers in the space under each question. Refer to the bedroom mirror, the matatu rear-view mirror and the letter 'A' beside a plane mirror. Use diagrams where they help, and use mathematical words such as mirror line, object distance, image distance, perpendicular, congruent and coordinates. Show your working when you calculate or use coordinates.

### Part 1: The puzzle and the mirror line

**Aisha stands in front of her full-length bedroom mirror in Nairobi. She raises one hand and her reflection raises the opposite hand. The Kanga skirt pattern and the buttons on her shirt all look the same. In class, your teacher placed a bold letter 'A' beside a plane mirror and the image looked flipped left to right.**

**(a) What is a line of symmetry, and how is it connected to a mirror line?**

**(b) From your Symmetry Safari, how many lines of symmetry did a circle, a square, a rectangle, an equilateral triangle and a scalene triangle have?**

**(c) Explain how the fold test and the mirror test both give the same lines of symmetry.**

(a) A line of symmetry divides a plane figure into two congruent halves that are exact mirror images of each other. It is also called a mirror line. A mirror line is the invisible line across which a figure is reflected, so a line of symmetry is the special case where the figure and its image are the same figure.

(b) Circle: infinitely many. Square: 4. Rectangle: 2. Equilateral triangle: 3. Scalene triangle: 0.

(c) When you fold a shape along a line of symmetry, the two halves fit exactly on top of each other. When you place a plane mirror on the same line, the half you see plus its reflection looks like the whole shape. Both tests give the same lines, so folding and reflecting are geometrically equivalent operations.

## Part 2: The three properties of reflection

**In Lesson 2 you measured the position of images in a plane mirror and constructed image points with a ruler and set square.**

**(a) State the three properties of reflection on a plane surface.**

**(b) Otieno stands 1.2 m in front of a plane mirror. How far is he from his image? Show your working.**

**(c) Use all three properties to explain why Aisha's bedroom-mirror image looks the same size and shape as her, and appears directly across the mirror from her, even though it is laterally reversed.**

(a) (1) The object distance from the mirror line equals the image distance from the mirror line. (2) The line joining an object point to its image is perpendicular to the mirror line. (3) The image is congruent to the object.

(b) The image is also 1.2 m behind the mirror, so the distance from Otieno to his image is  $1.2 \text{ m} + 1.2 \text{ m} = 2.4 \text{ m}$ .

(c) Property 3 says the image is congruent to the object, so the Kanga pattern, the number of buttons and her size are all preserved. Property 1 says the image is as far behind the glass as she stands in front. Property 2 says the line from her to her image meets the mirror at a right angle, so the image appears directly across the mirror and not off to one side. The image is congruent but laterally reversed, which is why it raises the opposite hand. The matatu driver's rear-view mirror follows the same three rules.

## Part 3: Reflection on the Cartesian plane

**On the Cartesian plane, the y-axis ( $x = 0$ ), the x-axis ( $y = 0$ ) and the line  $y = x$  can each act as a mirror line.**

**(a) Write the rule for reflecting a point  $(x, y)$  in each of these three lines.**

**(b) Point  $A(2, 5)$  is reflected in each of the three mirror lines. Write the coordinates of each image.**

**(c) Aisha stands at  $x = 3$  in front of a mirror wall along the y-axis. Where is her image, and how does this explain why the image raises the opposite hand?**

(a) Reflection in  $x = 0$  (the y-axis):  $(x, y) \rightarrow (-x, y)$ . Reflection in  $y = 0$  (the x-axis):  $(x, y) \rightarrow (x, -y)$ . Reflection in  $y = x$ :  $(x, y) \rightarrow (y, x)$ , so the two coordinates swap.

(b) In  $x = 0$ :  $A'(-2, 5)$ . In  $y = 0$ :  $A'(2, -5)$ . In  $y = x$ :  $A'(5, 2)$ .

(c) Her image is at  $x = -3$ , the same distance from the y-axis on the opposite side. Only the x (left–right) coordinate changes sign, while the y (up–down) coordinate stays the same. So left and right are flipped but height is preserved, which is why the image raises the opposite hand.

## Part 4: Finding a hidden mirror line

**In Lesson 6 you found the**

(a) Place the compass point on P and draw an arc that is more than half the length of PP'. With the same radius, draw an arc from

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| <p>mirror line when the mirror had been removed.</p> <p>(a) A point P and its image P' are given on plain paper with no mirror shown. Describe how you construct the mirror line with a ruler and compass.</p> <p>(b) Explain why the mirror line must pass through the midpoint of PP' AND be perpendicular to PP'. What would go wrong with any other line through the midpoint?</p> <p>(c) On the Cartesian plane, how can you find the mirror line for an object point and its image by using algebra? Include what happens when the segment joining them is horizontal or vertical.</p> | <p>P' so that the arcs cross above and below the segment. Join the two crossing points with a ruler. This line is the mirror line.</p> <p>(b) Property 1 says the object and image are equal distances from the mirror line, measured along the perpendicular from each point to the line. Property 2 says the segment PP' meets the mirror line at a right angle. Together these mean the mirror line cuts PP' at its midpoint and at 90°, so it is the perpendicular bisector of PP'. Suppose a line through the midpoint is not perpendicular to PP'. Then the perpendicular distances from P and P' to that line are smaller than the distance from each point to the midpoint, and the segment PP' is not perpendicular to the line. That line would break property 2, so reflecting P in it would give a different image from P'. Only the perpendicular bisector makes P' the true reflection of P.</p> <p>(c) Step 1: find the midpoint of P(<math>x_1, y_1</math>) and P'(<math>x_2, y_2</math>) as <math>((x_1 + x_2)/2, (y_1 + y_2)/2)</math>. Step 2: find the gradient m of PP' as <math>(y_2 - y_1)/(x_2 - x_1)</math>. Step 3: the mirror line is perpendicular to PP', so its gradient is the negative reciprocal, <math>-1/m</math>. Step 4: write the equation using <math>y - y_1 = m(x - x_1)</math>, with the midpoint as the point (<math>x_1, y_1</math>) and <math>-1/m</math> as the gradient. Special cases: if PP' is horizontal, the mirror line is vertical, x = the x-coordinate of the midpoint. If PP' is vertical, the mirror line is horizontal, y = the y-coordinate of the midpoint.</p> |
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### Part 5: Answering the driving question

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| <p><b>Why does a mirror show a perfect copy of the world, and where exactly does that copy live?</b></p> <p><b>Write a full explanation of about 6 to 8 sentences. Use the bedroom mirror, the matatu rear-view mirror and the letter 'A'. Include the unifying idea that connects the three properties of reflection. Give one real-life Kenyan application from Lesson 7 that works for the same reason.</b></p> | <p>A mirror shows a perfect copy because reflection is a congruence transformation, so the image has the same size and shape as the object, like the Kanga pattern and buttons on Aisha's skirt or the letter 'A'. The copy lives behind the mirror, exactly as far from the mirror line as the object is in front of it, because the mirror line is the perpendicular bisector of the segment joining every object point to its image. This one idea gives all three properties: equal distances, a joining line perpendicular to the mirror, and congruence. The image looks flipped because the mapping keeps distances along the mirror line the same but reverses the direction perpendicular to the mirror, which is why the letter 'A' looks laterally reversed and Aisha's reflection appears to raise the opposite hand. The matatu driver's rear-view mirror follows the same rules, so vehicles behind appear at a true proportional distance. The same rule works in a Kanga or kitenge design, where a motif is reflected across a mirror line, and in the symmetric front of the Parliament Buildings in Nairobi. The copy lives at the point where the perpendicular from the object to the mirror line, extended an equal distance beyond the mirror, ends.</p> |
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### RUBRIC

| Criterion                                           | Excellent (4)                                                                         | Proficient (3)                                                                  | Developing (1–2)                                                                 |
|-----------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Understanding of lines of symmetry and mirror lines | Defines a line of symmetry accurately, links it to the mirror line, gives all correct | Defines a line of symmetry and gives most counts correctly, with a partial link | Gives an incomplete definition or several incorrect counts, with no link between |

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|                                           | counts for the shapes and explains why the fold and mirror tests agree.                                                                                                                                                                                                                                                           | to the mirror line.                                                                                              | folding and mirroring.                                                                           |
| Properties of reflection and calculations | States all three properties precisely and correctly calculates 2.4 m, using the properties to explain the phenomenon fully.                                                                                                                                                                                                       | States the three properties and calculates the distance correctly, with a partial explanation of the phenomenon. | Names fewer than three properties or makes errors in the distance calculation.                   |
| Cartesian reflection rules                | Gives all three rules correctly, finds all three images correctly and explains the lateral flip through the sign change.                                                                                                                                                                                                          | Gives the rules and most images correctly, with a limited explanation of the flip.                               | Confuses the rules or gives incorrect image coordinates.                                         |
| Finding the mirror line and reasoning     | Describes the compass construction accurately, justifies the perpendicular bisector using equal distances and perpendicularity (showing why a non-perpendicular line fails), and gives the full algebraic method: midpoint, negative reciprocal gradient, $y - y_1 = m(x - x_1)$ , and the horizontal and vertical special cases. | Describes the construction and gives a partial justification or most steps of the algebraic method.              | Gives an unclear construction and no justification, or omits the perpendicular gradient.         |
| Final explanation using Kenyan contexts   | Gives a coherent, accurate explanation that links the perpendicular bisector to all three properties, the lateral reversal, the matatu mirror and a Kenyan application.                                                                                                                                                           | Gives an accurate explanation that covers most ideas with some contexts.                                         | Gives an explanation that is incomplete or has misconceptions, with few links to the phenomenon. |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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